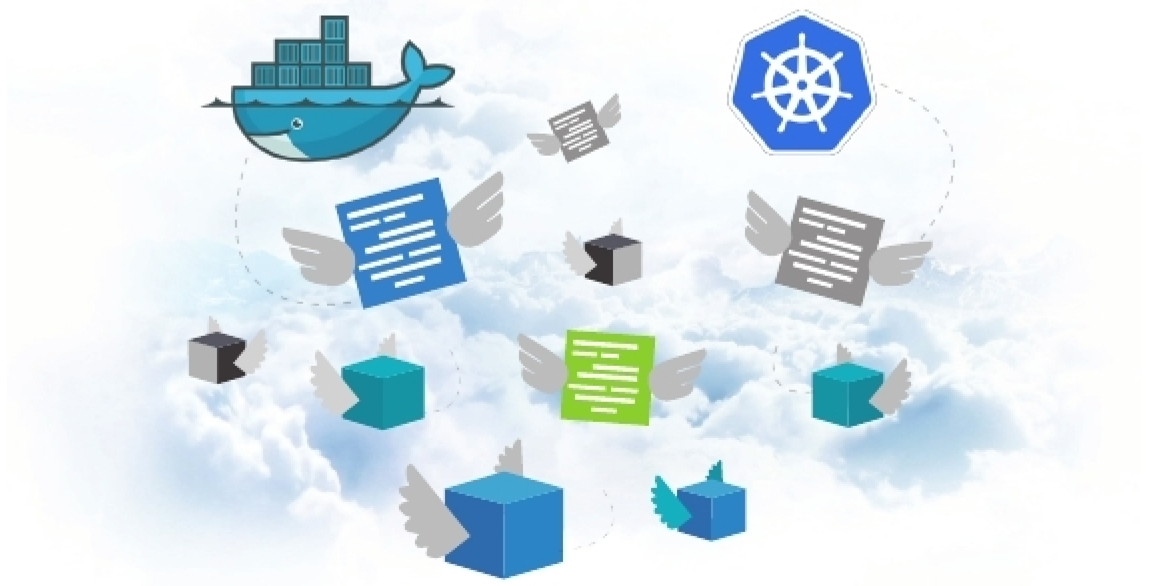
Migration of Application Workloads to Containers using Docker and Kubernetes

Containers present an easy way of deploying traditional applications, cutting-edge microservices and Big Data apps anywhere. In enterprises in which a large number of Docker pods may be in use, Kubernetes is one of the best orchestration tools to choose. Migrating application workloads to containers has many advantages for enterprises, such as profitability, scalability and security.



It was only a matter of time before containers changed the way enterprise cloud infrastructure was built and packaged. Containers have promoted and supported the concept of modularity, allowing applications and software to be independent of their life cycles. They're now used intensively with cloud orchestration and are therefore increasingly critical for enterprises looking to scale up and succeed with their cloud infrastructure.

Cloud orchestration

Orchestration refers to the process of automating the deployment of containers, which optimises applications. Container orchestration is imperative to manage the life cycles of containers in large environments. The most popular container orchestration tools today are Kubernetes, Marathon for Mesos and Docker Swarm. The configuration files point to the container images and establish networking between containers. These files are branched so that they can deploy the same applications across various environments, after which they are deployed to the production clusters.

Business advantages of the migration

The main reason for businesses to migrate to containers is that they offer a cost advantage. This is accomplished through a number of avenues.

- **Lower development costs:** Containers are portable and platform-independent, which eliminates the need to modify modules to make them compatible across various environments. This makes organisations more flexible, speeding up the development process.
- **Better security:** Since containers do not interact with each other, there are no security risks involved with any module crashing.
- **Smooth scaling:** Containers allow horizontal scaling. This allows you to run the containers you need in real-time, reducing your costs drastically.

Docker has been fairly successful, with almost every enterprise trying it out and more than 66 per cent of them adopting it thereafter. Its adoption continues to increase rapidly, with major corporations like PayPal, Uber, Spotify (which has more than 300 servers up and running) and even institutes like Cornell University choosing to adopt it. It is highly popular among most enterprises due to its support for multiple languages, and has a host of other benefits, such as:

- It reduces the infrastructure resources by a huge margin, assisting organisations in saving on a number of fronts, including server costs and employee costs.
- Container orchestration is already a speedy process, but Docker is able to deploy software and applications in seconds.

Kubernetes

Kubernetes is the popular open source orchestration tool for Docker containers and has grown rapidly over the past few years — not just as a tool, but even as a community, with more services, support and assisting tools now widely available. A number of cloud native Kubernetes distributions, like Azure Kubernetes Service, Google Kubernetes Engine, Amazon Elastic Kubernetes Service and DigitalOcean Kubernetes Service (DOKS) are being widely adopted to deploy and manage the containers.

Using cloud native Kubernetes distribution: The business advantages

The cloud native Kubernetes distribution has lightened the enterprise workload by a